

# DART Installation Guide

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This document provides detailed installation instructions for DART (Digital Asset Routing and Transformation) on macOS, Windows, and Linux systems.

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## Seeklight Credentials (Optional)

Seeklight is optional. For each colleague using Function 5, place a file named `seeklight_credentials.json` in their own user profile, **not** in the DART installation directory or distribution archive:

- macOS: `~/Library/Application Support/DART/seeklight_credentials.json`
- Windows: `%APPDATA%\DART\seeklight_credentials.json`
- Linux: `~/.config/DART/seeklight_credentials.json` (or `$XDG_CONFIG_HOME/DART/seeklight_credentials.json`)

Create the directory and file locally with these JSON fields, using the base URL and API key issued to that user or institution:

```
{
  "base_url": "https://YOUR-SEEKLIGHT-API-URL",
  "api_key": "YOUR-API-KEY"
}
```

On macOS/Linux, restrict the directory and file to your user after creating them (quote paths containing spaces):

```
chmod 700 "$HOME/Library/Application Support/DART"
chmod 600 "$HOME/Library/Application
Support/DART/seeklight_credentials.json"
```

On Linux, substitute the Linux configuration path above. On Windows, store the file in your private user profile with access restricted to your account. DART refuses a macOS/Linux credential file readable by

other users. The file is **plaintext**, not encrypted: do not email it, commit it, or put it into a DMG/ZIP. Anyone given a shared key can use that key, so request separate keys and rotation guidance from the Seeklight provider where possible. The build scripts exclude this filename, `.env*` files, and legacy `api-info/` folders from packages. Environment variables take precedence; the sibling `Seeklight-Resources/api-info/` files remain a development-only fallback.

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## macOS Installation (DMG)

DART for macOS is distributed as a DMG file with an automated installer that requires **no admin permissions**.

### Prerequisites

- **macOS 12** (Monterey) or later
- **Python 3.8+** must be installed
  - Download from: [python.org/downloads](https://python.org/downloads)
  - Or install via Homebrew: `brew install python`
  - Verify: `python3 --version`

### Installation Steps

#### 1. Download the DMG

- Download `DART_vX.x.dmg` from your distribution source

#### 2. Open the DMG

- Double-click the downloaded DMG file
- A new window opens showing the installer contents.

Note for Mac users who are UPDATING DART to a new version: You may need to remove or move your `~/DART/` folder to get it out of the way BEFORE the `Install DART.command` can run successfully!

#### 3. Run the Installer

- Double-click: `Install DART.command`

#### Note for first-time users (unsigned app) and updates:

- If the installer won't open, macOS Gatekeeper is blocking it
- Right-click on `Install DART.command` → select "Open"
- If the app is blocked close the dialog then proceed to `System Settings...` and `Privacy & Security`
- The key is to look for a statement like "Install DART.command was blocked..." and there should be an `Open Anyway` button beside it. Click that button to allow the install.
- This is only needed once

#### 4. Follow the Installation

- A Terminal window opens with the installer
- The installer checks for Python 3

- Confirms installation location: `~/DART/`
- Copies all files (takes a few seconds)
- Creates a launcher on your Desktop
- Press Enter when finished

## 5. Eject the DMG

- Right-click the mounted DART volume → Eject
- Or drag it to the Trash

## Installation Location

DART is installed to: `~/DART/` (your home directory)

```
~/DART/
├── Launch DART.command    # Double-click to start DART
├── run.sh                 # Root compatibility launcher
├── app.py                 # Main application
├── scripts/              # Launcher and helper scripts
│   └── run.sh             # Launcher script
├── common_dg_utilities/   # Utility functions (included)
├── INSTALLATION.md        # This file
├── README.md              # Full documentation
└── ... (other files)
```

## Desktop Shortcut

The installer creates: **Launch DART** on your Desktop

- Double-click this to launch DART at any time
- It's a symbolic link to `~/DART/Launch DART.command`

## First Launch

### 1. Start DART

- Double-click: **Launch DART** (on Desktop)
- Or navigate to `~/DART/` and double-click: **Launch DART.command**

### 2. Automatic Setup (first time only)

- A Terminal window opens
- Python virtual environment is created in `~/DART/.venv/`
- Dependencies install automatically (1-2 minutes)
- Progress is shown in the Terminal

### 3. DART Window Opens

- The DART application window appears when ready
- The Terminal window stays open (don't close it!)

- Minimize the Terminal if you want it out of the way

#### 4. Subsequent Launches

- Just double-click the launcher
- DART starts immediately (no setup needed)
- Dependencies are already installed

### Why No Applications Folder?

#### Traditional approach issues:

- Many work computers restrict Applications folder access
- Requires admin permissions to install
- Users encounter permission errors

#### DART's solution:

- Installs to `~/DART/` in your home directory
  - No admin permissions required
  - Works on all systems, including managed/restricted computers
  - Clean, predictable location
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## Windows Installation (ZIP)

DART for Windows is distributed as a ZIP archive.

### Prerequisites

- **Windows 10** or **Windows 11**
- **Python 3.8+** must be installed
  - Download from: [python.org/downloads](https://python.org/downloads)
  - **During installation: Check "Add Python to PATH"** 
  - Verify: Open Command Prompt, type `python --version`

### Installation Steps

#### 1. Download the ZIP

- Download `DART_v2.x_Windows.zip` from your distribution source

#### 2. Extract the Archive

- Right-click the ZIP file → Extract All...
- Choose a convenient location (e.g., `C:\Users\YourName\DART\`)
- Or extract to your Desktop

#### 3. Installation Location

- The extracted folder contains all DART files
- No further installation needed

## First Launch

### 1. Open the DART folder

- Navigate to where you extracted the files

### 2. Run the launcher

- Double-click: `run.bat`

### 3. Automatic Setup (first time only)

- A Command Prompt window opens
- Python virtual environment is created in `.venv\`
- Dependencies install automatically (1-2 minutes)
- Progress is shown in the console

### 4. DART Window Opens

- The DART application window appears when ready
- The Command Prompt window stays open (don't close it!)
- Minimize the console if you want it out of the way

### 5. Subsequent Launches

- Just double-click `run.bat`
- DART starts immediately

## Creating a Desktop Shortcut (Optional)

1. Right-click `run.bat` → Send to → Desktop (create shortcut)
  2. Rename the shortcut to "DART"
  3. Double-click the Desktop shortcut to launch DART
- 

## Linux Installation (Source)

DART runs from source on Linux systems.

### Prerequisites

- **Python 3.8+**
  - Most Linux distributions include Python 3
  - Verify: `python3 --version`
  - Install if needed:
    - Debian/Ubuntu: `sudo apt install python3 python3-venv`
    - Fedora/RHEL: `sudo dnf install python3`
    - Arch: `sudo pacman -S python`

### Installation Steps

#### 1. Download or Clone

```
# Clone from repository:
git clone https://github.com/Digital-Grinnell/DART.git
cd DART
```

## 2. Install common-DG-utilities (required dependency)

```
cd ..
git clone https://github.com/Digital-Grinnell/common-DG-utilities.git
cd DART
```

## 3. Run DART

```
./run.sh
```

## 4. First Launch

- The script creates a Python virtual environment in `.venv/`
- Installs all dependencies automatically
- Launches the DART application
- Subsequent runs skip the setup phase

**Note for developers:** The `python_requirements.txt` file includes `-e ../common-DG-utilities` which references the sibling repository. This line is automatically removed by the build scripts when creating DMG/ZIP distributions since those packages bundle the utilities directly.

## Creating a Desktop Launcher (Optional)

Create `~/local/share/applications/dart.desktop`:

```
[Desktop Entry]
Type=Application
Name=DART
Comment=Digital Asset Routing and Transformation
Exec=/path/to/DART/scripts/run.sh
Icon=utilities-terminal
Terminal=false
Categories=Utility;
```

Update `/path/to/DART/` with your actual path.

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## Prerequisites

Python 3.8 or Later

DART requires Python 3.8 or later. To verify your Python version:

```
# macOS/Linux
python3 --version

# Windows
python --version
```

## Installation:

- **macOS:**
  - Download: [python.org/downloads](https://python.org/downloads)
  - Or via Homebrew: `brew install python`
- **Windows:**
  - Download: [python.org/downloads](https://python.org/downloads)
  - **Important:** Check "Add Python to PATH" during installation
- **Linux:**
  - Usually pre-installed
  - If needed: `sudo apt install python3` (Debian/Ubuntu)

## Python Dependencies

The following Python packages are installed automatically on first launch:

- **flet 0.25.2** — UI framework
- **flet-desktop 0.25.2** — Desktop-specific features
- **cryptography** — Encrypted settings storage
- **azure-storage-blob 12.19.0+** — Azure Blob Storage integration
- **Pillow** — Image processing
- **PyMuPDF (fitz)** — PDF processing
- **pandas** — Data manipulation

**You don't need to install these manually** — the `run.sh` / `scripts/run.sh` / `scripts/run.bat` launchers handle everything.

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## Troubleshooting

### macOS Issues

#### "Install DART.command can't be opened"

**Cause:** macOS Gatekeeper blocking unsigned app

#### Solution:

1. Right-click on **Install DART.command**
2. Select "Open" from the menu
3. Click "Open" in the security dialog
4. Only needed once

### "Python 3 not found" error

**Cause:** Python not installed or not in PATH

**Solution:**

```
# Install Python via Homebrew
brew install python

# Or download from python.org
# Then verify:
python3 --version
```

### Installer says "DART is already installed"

**Cause:** Previous DART installation exists

**Solution:**

- Type **y** to overwrite the old installation
- Or type **n** to cancel and manually remove **~/DART/** first

### DART window doesn't appear

**Cause:** Terminal window was closed during setup

**Solution:**

1. Open Terminal
2. Navigate: **cd ~/DART**
3. Run manually: **bash run.sh**
4. Watch for error messages

## Windows Issues

### "python is not recognized"

**Cause:** Python not installed or not in PATH

**Solution:**

1. Install Python from [python.org](https://python.org)
2. **During installation: Check "Add Python to PATH"**
3. Restart Command Prompt

4. Verify: `python --version`

### Dependencies fail to install

**Cause:** Network issues or pip outdated

**Solution:**

```
# Upgrade pip
python -m pip install --upgrade pip

# Then re-run DART
scripts\run.bat
```

### Virtual environment errors

**Cause:** Corrupt virtual environment

**Solution:**

```
# Delete the virtual environment
rmdir /s .venv

# Re-run launcher (recreates .venv)
scripts\run.bat
```

### Linux Issues

**`./run.sh`: Permission denied**

**Cause:** Script not executable

**Solution:**

```
chmod +x run.sh scripts/run.sh
./run.sh
```

**"No module named 'tkinter'"**

**Cause:** tkinter not installed (required by flet)

**Solution:**

```
# Debian/Ubuntu
sudo apt install python3-tk
```

```
# Fedora/RHEL
sudo dnf install python3-tkinter
```

## Common Issues (All Platforms)

### DART closes immediately after launch

**Cause:** Error during startup

**Solution:**

1. Look for error messages in the Terminal/Console window
2. Check log files in `~/DART-data/logfiles/` or `{working_folder}/logfiles/`
3. Common causes:
  - Missing Python dependencies (re-run installer)
  - Corrupted settings file (delete `~/DART-data/persistent.json`)

### "common\_dg\_utilities not found" error

**Cause:** DMG/ZIP was missing the utilities folder

**Solution:**

- This should not happen with properly built installers v2.2+
- The build scripts automatically bundle common\_dg\_utilities
- For developers: Ensure `../common-DG-utilities/` exists when building

### "not a valid editable requirement" error

**Cause:** Using source repository with missing sibling common-DG-utilities repo

**Solution (for developers):**

```
# Clone the required utilities repository
cd ..
git clone https://github.com/Digital-Grinnell/common-DG-utilities.git
cd DART
```

**Note:** DMG/ZIP installers automatically remove this dependency since utilities are bundled directly.

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## Uninstallation

### macOS

1. Delete the application folder:

```
rm -rf ~/DART
```

## 2. Delete user data (optional):

```
rm -rf ~/DART-data
```

## 3. Remove Desktop shortcut:

- Drag **Launch DART** from Desktop to Trash

## Windows

### 1. Delete the DART folder:

- Navigate to where you extracted DART
- Delete the entire folder

### 2. Delete user data (optional):

- Navigate to: **C:\Users\YourName\DART-data\**
- Delete the folder

### 3. Remove Desktop shortcut (if created):

- Delete the DART shortcut from Desktop

## Linux

```
# Delete DART folder
rm -rf ~/path/to/DART

# Delete user data (optional)
rm -rf ~/DART-data

# Remove desktop launcher (if created)
rm ~/.local/share/applications/dart.desktop
```

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## Support

### Documentation

- **README.md** — Full application documentation
- **QUICKSTART.md** — Quick reference guide
- **Function Help** — Built-in help for each function (Help Mode toggle in app)

### Getting Help

- **Repository:** <https://github.com/Digital-Grinnell/DART>
- **Issues:** Report bugs on GitHub Issues
- **Questions:** Use GitHub Discussions

## Log Files

DART maintains detailed log files for troubleshooting:

- **Startup logs:** `~/DART-data/logfiles/`
- **Operational logs:** `{working_folder}/logfiles/` (set in app)
- **Format:** `dart_YYYYMMDD_HHMMSS.log`

Include relevant log excerpts when reporting issues.

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## System Requirements

### Minimum

- **OS:** macOS 12+, Windows 10+, or modern Linux
- **Python:** 3.8 or later
- **Memory:** 2 GB RAM
- **Disk:** 500 MB free space
- **Display:** 1280x720 or higher

### Recommended

- **OS:** macOS 13+, Windows 11, or latest LTS Linux
  - **Python:** 3.10 or later
  - **Memory:** 4 GB RAM
  - **Disk:** 1 GB free space
  - **Display:** 1920x1080 or higher
- 

## Security Note

DART stores sensitive settings (like Azure connection strings) using encrypted storage:

- **Encryption key:** `~/DART-data/encryption_key`
- **Settings file:** `~/DART-data/persistent.json` (sensitive fields encrypted)
- **Working settings:** `{working_folder}/dart_settings.json` (per-project)

**Keep your encryption key secure.** If lost, you'll need to re-enter encrypted settings.

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*DART Version: 2.x*

*Documentation: <https://github.com/Digital-Grinnell/DART>*